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Constructive Processing in Children's Event Memory

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To examine effects of general event knowledge on specific event memories, nursery-school and kindergarten children participated in either 1 creative movement workshop (the *episodic condition*) or 4 workshops (the *repeated condition*), and recalled the workshops 4 weeks later. One half of the children also recalled the workshops on the same day. It was predicted that recall in the repeated condition would be more complete, better organized, but less accurate than in the episodic condition because of constructive processing. Results confirmed that (a) children recalled more from the first workshop (but not the last) in the repeated condition, (b) kindergarten children's recall in the repeated condition was better sequenced, and (c) more intrusions were reported in the repeated condition. Furthermore, (d) children recalled more activities with age, and (e) recall practice and specific cues increased amount, but not accuracy, of recall.

Children's memory for real-world events has emerged as an important area of developmental research for two reasons. First, a widespread concern that laboratory studies may underestimate young children's abilities has caused many researchers to develop more naturalistic and ecologically valid methods for studying cognitive development (DeLoache, Cassidy, & Brown, 1985, Donaldson, 1978, Nelson, 1986). The study of children's memory for real-world events offers a window onto how children naturally organize their knowledge of the world and what they remember from their own experiences. Second, the question of whether young children can accurately recall real-world events has become a critical issue in evaluating children's legal testimony (Ceci, Ross, & Toglia, 1987). In particular, there is a need to find out how various factors such as familiarity, participation, emotional salience, procedures for elicitation, and the nature of the memory response affect children's event recall at different ages.

Familiarity with an event has emerged as an important factor influencing both how much children remember about specific episodes and whether what they recall is accurate. Hudson and Nelson (1986) found an increase in general information reported and a decrease in episodic information reported in children's recall of specific episodes as a function of increased experience with events, suggesting that specific episodes were confused with general event knowledge. Likewise, Myles-Worsley, Cromer, and Dodd (1986) found that, with increased school experience, elementary school-age children were less accurate in their recall of preschool activities and tended to confuse their current school experiences with memories of nursery school. In contrast, novel or unusual events are recalled more accurately. Hudson and Fivush (in press, Fivush, Hudson, & Nelson, 1984) found no decrease in accuracy in children's memory for an unusual kindergarten field trip over a period of 6 years, and Sheingold and Tenney (1982) found that children's memories of the birth of a sibling were accurate up to 8 years later. Comparing children's memory for familiar events (birthday parties and library visits) to their recall of a more novel event (a visit to a museum of natural history), Todd (1986) found that there were more errors made in children's recall of the more familiar events.

These findings indicate that increased experience with an event leads to greater general knowledge about that event, but also results in more distortions in recall as compared with memory for novel events. Similar effects are found in children's memory for stories about events. General knowledge, organized as event schemas or scripts, increases the quantity and organization of script-based stories but also distorts memory in the direction of a canonical script (Hudson, 1988, Hudson & Nelson, 1984, McCartney & Nelson, 1981). These types of distortions are characteristic of schematic or constructive processing in memory and indicate that general event schemas are guiding recall (Bartlett, 1932, Mandler, 1983, 1984). This line of research has also found consistent developmental differences in effects of constructive processing; preschool children tend to rely more heavily on general knowledge structures than do older children and adults. They remember stories about familiar events better than stories about less familiar events, and they are less able to accomodate script deviations in their story recall (Hudson & Nelson, 1984).

Do age differences in constructive processing generalize to children's memory for events? Unfortunately, comparisons of older and younger children's memory for novel versus familiar events are confounded because the novel events studied (e.g., birth of a sibling) are very different from the familiar events

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studied (e.g., nursery school) along many dimensions other than just familiarity (e.g., emotional salience, degree of participation, complexity, and duration). Variations in methodology may also contribute to observed differences in amount and accuracy of recall. Hudson and Nelson (1986) used only general elicitation questions; Todd (1986) used both general questions and specific cues; Hudson and Fivush (in press, Fivush et al., 1984) used general questions, specific probes, and photograph cues; Sheingold and Tenney (1982) used a lengthy questionnaire, but did not provide cues; and Myles-Worsley et al. (1986) relied on picture recognition and sequencing measures. Finally, retention interval and amount of recall practice also vary considerably across these studies.

One recent study that controlled for event characteristics and retention interval did find developmental differences in effects of general event knowledge on recall of specific episodes. Farrar and Goodman (in press) examined 4- and 7-year-olds' memory for a deviant episode of a repeated experience. Children visited a laboratory playroom on four occasions; one occasion differed from the other three. The younger children's recall of the deviant episode was facilitated by prior experience with the event, and they were more likely to confuse episodes. The older children formed distinct memories for both the deviation episode and the general event script and relied less on general event knowledge in recalling specific episodes. Whether developmental differences are also found in memory for more typical events is the focus of the present study.

Children's memory for novel versus familiar events was examined by comparing children's memory for the same events experienced as either single episodes or as one of a series of similar episodes. Nursery-school and kindergarten children participated in either a single creative movement workshop (the *episodic condition*) or four workshops (the *repeated condition*), and recalled specific workshops 1 month later. In this way, amount of experience was manipulated while event characteristics were held constant and retention interval was controlled. It was predicted that children in the repeated condition would form general event representations of the event over four episodes, which would be used to organize their long-term recall. Reliance on event schemas should increase the amount and organization of children's recall but decrease the accuracy of their recall. If younger children's recall is more dependent on general event knowledge than older children's, effects should be greater for the nursery-school children.

Effects of recall practice and cues were also examined by asking one half of the children to recall the workshops on the same day and by giving children specific verbal prompts and photograph cues in delayed recall. Effects of recall practice and cuing on both quantity and accuracy of recall were assessed. For example, immediate recall could help preserve memory for details of individual episodes, resulting in fewer distortions in delayed recall. In addition, although specific questionning increases the amount of information children recall, it can also increase inaccuracies in recall (Dent & Stephenson, 1979).

Method

Design

The schedule for the workshops and recall sessions is shown in Table 1. A total of 20 children in each age group participated in a series of weekly creative movement workshops for 4 weeks (the repeated condition). One week after the last workshop (Week 5), children were asked to recall the first workshop, and 3 weeks later (Week 8), they were asked to recall the last workshop. A total of 20 children in each age group also participated in a single workshop, Workshop A, identical to the first workshop in the repeated condition, and 20 children in each age group participated in a single workshop, Workshop D, identical to the last workshop in the repeated condition (the episodic condition). Children in the episodic condition were also asked to recall the workshops 4 weeks later. Because of absences, only 19 of the kindergarten children were interviewed in the delayed recall session in both episodic conditions.

Nursery and kindergarten classes were randomly assigned to either the repeated condition or one of the two episodic conditions. Within each class, equal numbers of children were randomly assigned to either practice or control groups. Children in the control groups only recalled the workshops in delayed recall, but children in the practice groups were also asked to recall the workshops on the same day as the workshops took place. Because the practice interviews focused on the particular activities of that day's workshop, immediate recall of the first workshop was not expected to affect immediate recall of the last workshop. Likewise, delayed recall of the first workshop was not expected to affect delayed recall of the last workshop. These assumptions were tested by comparing immediate and delayed recall of the first and last workshops by children in the repeated condition.

Subjects

Subjects were 60 children from nursery school (M age = 4 years, 7 months; age range = 3 years, 6 months–5 years, 1 month) and 58 children from kindergarten (M age = 5 years, 8 months; age range = 5 years, 2 months–6 years, 2 months), who attended public and private preschool programs that drew from mixed ethnic and socioeconomic populations in New York City. The children's academic ability ranged from low average to gifted. Approximately equal numbers of boys and girls participated in the study.

Procedure

Workshops. An instructor, who was unknown to the children, came to their classrooms and conducted the workshops. Workshops included familiar types of activities such as singing, dancing, and movement games, but most of the specific songs, dances, and games were novel and the particular sequence of activities was unfamiliar. Each workshop followed the same format: Everybody sat in a circle and said "hello" while passing around a drum; then they sang a warm-up song. Next, two songs with accompanying movements were sung; specific songs varied in each workshop. The instructor then led the children in whole-body activities and dances that also varied across workshops. Finally, they returned to the circle, did a relaxation exercise, and said "goodbye." Each workshop lasted approximately 25–30 min.

Practice interviews. Children in the practice groups were interviewed individually by one of two female experimenters who each interviewed equal numbers of children. Children were first asked, "What happened in the workshop today?" This question was followed by *general prompts* ("Anything else?" "What was the first thing that happened?"). Next, children were asked *specific probes* such as "What songs did you sing?" to remind them of any activities that they failed to report in response to the general prompts.

Delayed recall interviews. Every child was interviewed individually by one of the experimenters who conducted the practice interviews. Children in the practice groups were interviewed by different experimenters in the practice and delayed recall interviews. Children in the repeated condition were asked, "What happened in the first (or last) workshop?" and children in the episodic condition were asked, "What happened in the workshop?" These questions were followed by the same general prompts and specific probes used in the practice interviews.

Table 1
Schedule of Workshops and Recall Sessions

	Workshops				Recall sessions	
Condition	Week 1: A	Week 2: B	Week 3: C	Week 4: D	Week 5: A	Week 8: D
Repeated						
Practice	X[a]	X	X	X[a]	X	X
Control	X	X	X	X	X	X
Episodic (A)						
Practice	X[a]				X	
Control	X				X	
Episodic (D)						
Practice				X[a]		X
Control				X		X

[a] Practice recall session held on the same day as the workshop.

Next, children were shown five or six photographs showing various workshop activities.[1] Children were asked if they could remember what was happening in the picture. If children merely described the photograph (e.g., "They're sitting in a circle"), they were asked additional probe questions to test whether they actually recalled the activity.

Coding

Practice and delayed interviews were audio tape-recorded and transcribed verbatim for coding. In total, there were 10 specific activities for each workshop that could be recalled: saying hello, the two songs, the four whole-body activities, a dance, the relaxation exercise, and saying good-bye. Children were credited with recall of a specific activity if they mentioned it in their recall in such a way that it could not be confused with any other activity (e.g., "We sang Rum Sum Sum"). The type of cue (general, specific, photograph) that elicited recall of each activity was also coded. Mere descriptions of photographs were not included as recall: Children had to name the specific activity depicted in the photograph.

Any activity mentioned that had not occurred in the specific workshop under discussion was coded as an *intrusion.* It was expected that children in the repeated condition would produce intrusions based on recalling activities from the other workshops. However, intrusions could also occur in the episodic condition, particularly if children were asked specific probes such as "What song did you sing?" Rather than answer "I don't know," children might respond with the name of any song. Thus, the intrusion rate in the episodic condition served as a baseline to compare with the rate of intrusions in the repeated condition.

If children referred to activities at a more general level, such as "We sang songs" or "We did exercises," these were coded as *generalizations.* Intrusions and generalizations were not included in the analyses of amount recalled, but were analyzed separately.

Two coders scored a sample of 20% of the recall protocols, representing a cross section of protocols from all ages, groups, and conditions. Intercoder agreement was 90%. One coder coded the remaining protocols.

Results

Children's recall in the repeated condition was analyzed using a 2(age: kindergarten and nursery school) × 2(group: practice and control) × 2(workshop: first and last) × 3(cue: general, specific, and photograph) mixed-model analysis of variance, with age and group as between-subjects factors and workshop and cue as within-subjects factors. To compare recall across conditions, separate 2(condition: repeated and episodic) × 2(age) × 2(group) × 3(cue) analyses were performed for Workshop A and Workshop D, with condition, age, and group as between-subjects factors and cue as a within-subjects factor.

When appropriate, a secondary analysis of Workshops A and D in the episodic conditions was performed with age, group, and workshop as between-subjects factors and cue as a within-subject factor. This analysis was used to test whether workshop effects were due to difference in recall of the first and last workshops in a series (only apparent in the repeated condition), or to an inherent difference in the memorability of the specific activities from each workshop (also apparent in the episodic condition). Because the focus of the study was not on the inherent differences in the two workshops—they were designed to be equally memorable—this analysis was used only to clarify workshop effects in the repeated condition analysis.

Finally, to test for changes in amount recalled over time, a secondary analysis compared children's immediate recall in the practice interviews with their delayed recall. For this analysis, interview (practice and delayed) was added as a within-subjects factor; group was eliminated as a factor; and the cue factor only had two levels, general and specific, with responses to specific and photograph cues collapsed in delayed recall measures.

The significance level was set at $p < .01$ for all effects, and Newman-Keuls post hoc tests at $p < .05$ were performed for all interaction effects.

Amount Recalled

The mean numbers of correct activities (out of 10) recalled by cue in immediate recall for children in the practice groups are displayed in Figure 1. Figure 2 shows the same data for all children in delayed recall, as well as the mean number of generalizations and intrusions reported.

[1] Because photographs were not available for all of the classes, approximately one half of the children viewed photographs of their own class, and the remaining children viewed photographs of children in a different class of same-age children. A preliminary analysis showed no difference between the groups in their ability to identify the activities depicted in the photographs.

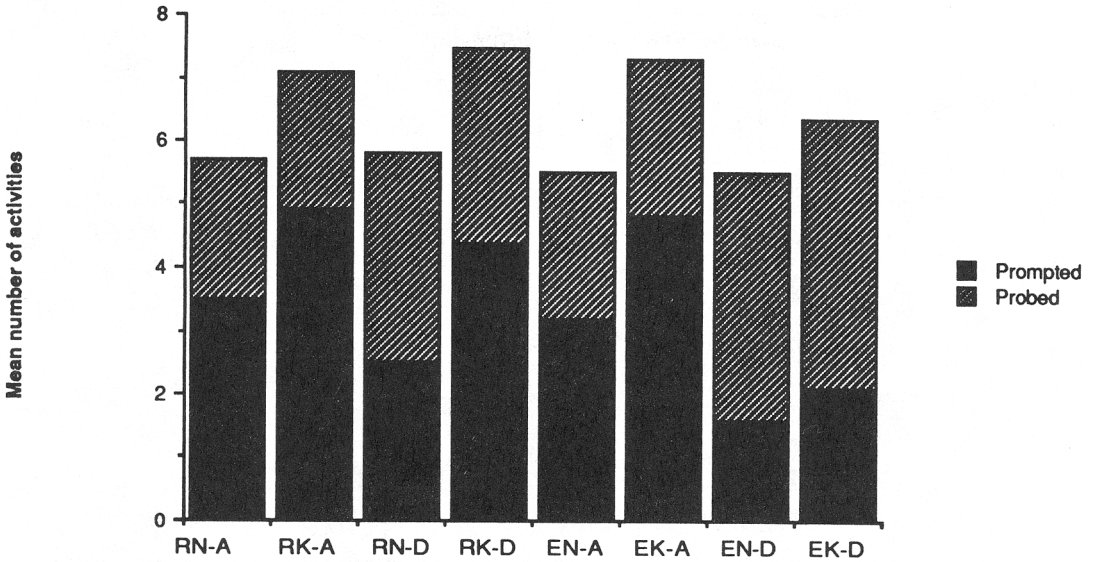


Figure 1. Mean number of activities reported by cue in practice recall. (R = repeated, E = episodic, N = nursery school, K = kindergarten, A = workshop A, and D = Workshop D.)

Delayed recall. An effect of age, $F(1, 36) = 8.3$, in the repeated condition showed that kindergarten children recalled more activities than did nursery-school children. Effects of workshop, $F(1, 36) = 9.29$, and cue, $F(2, 72) = 9.87$, and the interaction of these factors, $F(2, 72) = 8.11$, indicated that children recalled less in response to general cues in Workshop D than in Workshop A. There was also a practice effect, $F(1, 36) = 6.86$; children in the practice groups recalled more activities ($M = 5.79$) than did children in the control groups ($M = 4.36$).

The analysis of Workshop A revealed effects of age, $F(1, 71) = 25.91$, practice, $F(1, 71) = 11.99$ ($M = 5.75$ and 4.41 for control and practice groups, respectively), and cue, $F(2, 142) = 23.03$, in the same direction as that in the repeated condition. The Cue × Age interaction, $F(2, 142) = 7.99$, indicated that age differences were only apparent in children's responses to general prompts. There was also an effect of condition, $F(1, 71) = 9.19$; children in the repeated condition recalled more than children in the episodic condition.

Analysis of Workshop D produced the same practice effect, $F(1, 71) = 7.45$ ($M = 4.98$ and 3.67 activities for control and practice groups, respectively), but there were no other significant effects.

Because there was an effect of workshop in the repeated condition, an analysis of recall across both episodic conditions was also performed. Effects of age, $F(1, 70) = 12.52$, and practice, $F(1, 70) = 9.5$ ($M = 4.99$ and 3.67 activities for control and practice groups, respectively) replicated previous analyses. There was also an Age × Workshop interaction effect, $F(1, 70) = 7.1$, as well as an Age × Workshop × Cue interaction, $F(2, 140) = 4.35$. As shown in Figure 2, kindergarten children who participated in Workshop D recalled less in response to general prompts than did kindergarten children who participated in Workshop A; and recall in response to specific probes was greater for Workshop D than for Workshop A for both age groups.

Thus, children in the repeated condition recalled more from the first workshop than did children who participated only in a single workshop; older children tended to recall more than younger children, especially in response to general prompts; and all children regardless of age and experience recalled more in delayed recall if they had also recalled the workshop once before. Without specific cues, children in the repeated condition remembered more from Workshop A than from Workshop D, but even in the episodic condition, kindergarten children had difficulty recalling Workshop D with only general prompts.

All effects of condition, workshop, and practice involving children in the repeated condition were additive, not catalytic. That is, the effect of practice was no greater for children in the repeated condition who were exposed to both repeated experiences and repeated recall sessions than for children in the episodic conditions who had only one prior experience with the workshop and with recalling the workshop.

Immediate and delayed recall: Practice groups. By comparing the immediate and delayed recall, it is possible to test whether effects found in delayed recall were a result of constructive processes occurring over time or whether they reflected differences in recall of the workshop apparent on the same day the workshop took place. (Responses to specific probes and to photographs in delayed recall were collapsed into the level of specific cue for comparison to children's responses to specific probes in immediate recall).

The only significant effect in the analysis of the repeated condition was the Time × Cue interaction, $F(1, 19) = 13.31$, indicating that the amount of recall in response to general prompts decreased from immediate to delayed recall (see Figures 1 and 2).

The same interaction was found in analysis of Workshop A, $F(1, 36) = 14.09$, as well as effects of age, $F(1, 37) = 13.55$, and cue, $F(1, 37) = 8.63$, indicating that older children recalled more than younger children and that recall was greater in response to general prompts.

The analysis of Workshop D found that recall decreased over time, $F(1, 36) = 12.72$, and more actions were reported in response to general prompts, $F(1, 36) = 16.58$. There was also a

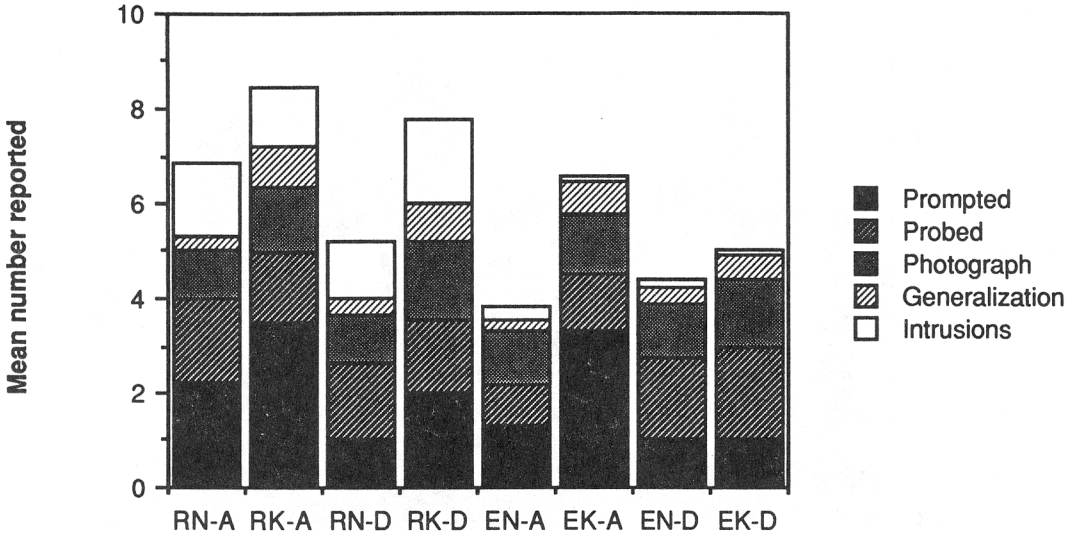


Figure 2. Mean number of activities by cue, intrusions, and generalizations reported in delayed recall. (R = repeated, E = episodic, N = nursery school, K = kindergarten, A = Workshop A, and D = Workshop D.)

Condition × Cue interaction, $F(1, 36) = 10.1$, such that children in the repeated condition recalled more in response to general prompts than did children in the episodic condition.

Finally, an analysis of recall in the episodic conditions found that recall decreased over time, $F(1, 35) = 14.94$, and that the number of activities reported in response to general prompts was greater for Workshop A than for Workshop D, $F(1, 35) = 26.92$.

Thus, children in the repeated condition remembered as much from the first and last workshops in immediate recall. However, Workshop D was somewhat more difficult to recall without specific cues even in immediate recall for children in the episodic condition. The whole-body activities in Workshop D (e.g., touching your chin with your knees) may have been more difficult to describe than those in Workshop A (e.g., make a round shape). Moreover, the shape movements in Workshop A were repeated more than once in the workshop, whereas the movements from Workshop D were only performed once in the workshop.

Also of interest is the finding that children in the repeated condition reported more activities from Workshop D in response to general prompts than did children in the episodic condition. This suggests that repeated experience with the event helped children to report more activities even in immediate recall. Otherwise, immediate recall was very similar across conditions (see Figure 1).

Intrusions and Generalizations

Overall, children in the repeated condition produced a mean 1.54 intrusions in delayed recall, whereas children in the episodic condition produced a mean of only 0.36. There were no significant effects in the analysis of the repeated condition. Condition effects were found in the analyses of both Workshop A, $F(1, 71) = 43.56$, and Workshop D, $F(1, 71) = 24.18$, and there were no other effects for either analysis.

Because there were virtually no generalizations reported in response to photograph cues, this cue level was not included in the analyses of frequency of generalizations (i.e., only two levels of the cue factor, general probes, and specific prompts were included). Age effects were found in the analysis of the repeated condition, $F(1, 36) = 6.13$, and in Workshop A, $F(1, 71) = 9.09$; kindergarten children produced more generalizations than did nursery-school children. In addition, cue effects were found in the analyses for the repeated condition, $F(1, 36) = 20.84$, and Workshop D, $F(1, 71) = 27.74$; more generalizations were reported in response to the general prompts. (See Figure 2.)

Because the amount of correct recall varied by age, younger children may have produced proportionally more intrusions and generalizations than older children. To test for this possibility, proportion scores for intrusions and generalizations were computed by adding the total number of correct activities, intrusion, and generalizations reported by cue, and dividing that sum by the number of intrusions and by the number of generalizations reported. Proportions were converted by arcsine transformation for analysis. (Again, only two levels of the cue factor, general and specific, were used in the analysis of proportion of generalizations.)

There were no significant effects in the analysis of proportion of intrusions in the repeated condition. In recall of Workshop A, a condition effect, $F(1, 71) = 13.41$, indicated that proportionally more intrusions were reported in the repeated condition ($M = .17$) than in the episodic condition ($M = .04$). Analysis of Workshop D also produced a condition effect, $F(1, 71) = 15.36$, in the same direction ($M = .16$ and .05 for the repeated and episodic conditions, respectively), as well as an Age × Practice interaction, $F(1, 71) = 7.10$. Practice decreased the proportion of intrusions reported by kindergarten children ($M = .15$ and .08 for practice and control groups, respectively), but not nursery-school children ($M = .08$ and .13 for practice and control groups, respectively).

Table 2
Frequencies (and Percentages) of Correct and Incorrect Sequences by Age and Condition

Age/Condition	Correct	Incorrect	Total
Nursery-school			
Episodic	17 (89)	2 (11)	19
Repeated	30 (94)	2 (6)	32
Kindergarten			
Episodic	20 (80)	5 (20)	25
Repeated	35 (88)	5 (12)	40

There were no significant effects in the analysis of proportion of generalizations in the repeated condition. The interaction of condition and rehearsal, $F(1, 71) = 6.33$, found in the analysis of Workshop A indicated that children in the repeated condition who participated in the practice recall session produced proportionally fewer generalizations ($M = .07$) than did children in the control groups ($M = .17$). A cue effect, $F(1, 71) = 20.45$, was found in the analysis of Workshop D, so that proportionally more generalizations were produced in response to general prompts ($M = .28$) than to specific probes ($M = .02$).

These findings indicate that children tended to describe the workshops at a more general level when given only prompts, but provided more specific descriptions of activities in response to probe questions. Specific questioning produced both more specific responses and more intrusions, but the proportion of intrusions in response to the specific questions was no greater than for general prompts. There were no age differences in frequencies of intrusions nor in any of the proportion measures.

Sequence

Only interviews in which children reported more than one activity in response to general prompts could be used in the sequence analysis so that order of mention would not be confused with order of prompting. Because these frequencies were fairly low, each interview sequence was simply categorized as correct or incorrect. To be considered incorrect, at least one act was recalled out of order. Separate chi-square tests for each age group were performed on the number of children producing correct and incorrect sequences across conditions (summed over rehearsal groups). These data are displayed in Table 2.

Although sequencing was generally accurate in both conditions, more correct sequences were produced by children in the repeated condition. Differences in correct and incorrect sequences were found in the expected direction for both age groups, but only the kindergarten test was significant, $\chi^2(1) = 7.76$. $p < .05$. One reason for nursery-school children's apparent correct sequencing was that they seldom recalled more than one internal activity in either condition, but frequently recalled one activity on their own and then recalled another when asked "What was the first thing that happened?" or "What happened at the very end?" This sequence was considered correct, but children did not actually recall the actions in sequence. In contrast, for kindergarten children who recalled more actions independently, children in the repeated condition were more likely to sequence the activities in the correct temporal order.

Discussion

The focus of this investigation was whether children engage in constructive processing when recalling typical everyday events. The evidence is that they do. In delayed recall, children recalled more about an episode if they had experienced additional similar episodes instead of just one; but with increasing experience, children tended to confuse details of particular episodes. In addition, kindergarten children were better able to report a repeated event in temporal sequence than they were the one-time event. These characteristics of recall—exhaustiveness, and organization at the expense of intrusions—are hallmarks of constructive memory processes (Barlett, 1932, Mandler, 1983, 1984).

Compare these results with research on children's memory for script-based stories (Hudson & Fivush, 1983, Hudson & Nelson, 1984, McCartney & Nelson, 1981). Results are similar: Increased familiarity with an event results in more exhaustive memory, more sequentially organized recall, and more intrusions. The notable difference is the lack of age differences in effects of familiarity found in this study. As was found in previous studies of children's memory for real-world events (Hudson & Nelson, 1986), there was no evidence that younger children's memory was more schematic or script-dependent than that of older children. This suggests that age differences in constructive memory processes are more apparent in memory for stories about events and other types of prose (Mandler, 1984) than in recall of actually experienced events.

Yet Farrar and Goodman (in press) did find that 4-year-olds' memory for a deviant episode was more script-dependent than that of 7-year-olds. In particular, the younger children had greater difficulty recalling a deviant episode if it was the first of four events than if it was the last. In contrast, the present study found that without specific memory cues, children in the repeated condition recalled more from Workshop A than from Workshop D. However, kindergarten children in the episodic condition also recalled more from Workshop A than from Workshop D in free recall, indicating that Workshop A may have been easier to recall verbally, independent of familiarity.

This discrepancy in research findings is relevant to the issue of whether general event knowledge is a prerequisite for remembering specific episodes. First, consider some of the methodological differences between the two studies. It is possible that the procedure used in Farrar and Goodman's (in press) study (i.e., children were brought to an unfamiliar laboratory, then participated individually in a sequence of novel activities) made the event more novel. In contrast, the event studied in this experiment was more similar to a naturally occurring everyday event. This is not to say that the creative workshop event was necessarily more ecologically valid than the laboratory playroom event—each is similar to and different from many types of events that young children typically encounter. Rather, there may be important differences in how children attend to, participate in, and remember different kinds of events.

Another difference between the two studies is that Farrar and Goodman (in press) compared recall of 4- and 7-year-olds, and this study examined a younger age range, 3 to 6 years. In Hudson and Fivush's (1983) study of preschool children's recall of script-based stories, developmental differences were found

comparing 3- and 5-year-olds' recall. However, developmental changes in event memory may occur later than at 6 years of age.

Most important, Farrar and Goodman's (in press) study examined recall of deviant episodes, not ones that conformed to the general event structure as in this investigation. Their finding that 4-year-olds' recall of the deviation episode when it was occurred as the last episode was as lengthy and as accurate as that of 7-year-olds, supports the interpretation that developmental differences only occur when initial episodes are very different, not when they follow the same schema as in this study. Developmental differences in the relation between scripts and episodic memories may be more apparent when conditions make it difficult to represent episodes as instantiations of a common event schema.

These results also speak to issues of accuracy of children's memory for real-world events and the appropriate procedures for eliciting children's recall. Research on constructive memory processes in young children and children's eyewitness memory has produced conflicting results. On the one hand, there is evidence that preschool children are more suggestible in their recall than are older children (Ceci et al., 1987, Dale, Loftus, & Rathbun, 1978). On the other hand, research on constructive memory indicates that the ability to draw elaborative inferences, including distortions, increases from preschool to early elementary school years, suggesting that older children's memory is less accurate than that of younger children (Ceci, Caves, & Howe, 1981, Landis, 1982).

In light of past research, two distinctions need to be considered in order to make sense of the findings from this study. First, note that studies of suggestibility and inferential processing have primarily used stories, pictures, and films. As was discussed earlier, age differences found in recall of presented stimuli are less apparent in recall of naturally occurring events. Second, the probe questions used in this study did not include any misleading information or ask for any inferences on the part of the children. Although the issue of suggestibility in children is still under debate (cf. Goodman, Rudy, Bottoms, & Aman, in press, Loftus & Davies, 1984), the findings from this study show that when children are recalling events that they themselves have experienced and are not given any misleading information, nursery-school children's memory is as accurate as that of kindergartners. However, both age groups are likely to confuse memories of similar episodes. This tendancy is not limited to young children; general event schemas can distort memory for specific episodes at all ages (Linton, 1982, List, 1986, Neisser, 1981).

Results also indicated that children's response to general prompts—what could be considered their *free recall*—was very reliable. Although children reported more generalizations in free recall as opposed to cued recall, they also tended to recall more specific activities. Furthermore, the intrusion rate did not vary by cue: Children reported intrusions in both free and cued recall, and there was no evidence that more specific questioning produced more intrusions than were evident in their free recall.

This finding is important for developing age-appropriate interview techniques for eliciting children's recall of events. A consistent finding in memory research is that younger children spontaneously recall less than older children. Many age effects in amount recalled in this study could be attributed to the fact that older children reported more activities in free recall than did younger children. Given concerns about young children's suggestibility, one might be wary of using specific questions to elicit further recall from young children. These results suggest that specific questioning is appropriate for eliciting recall from preschool children and does not necessarily distort their recall of events they have experienced themselves.

Finally, these results indicate that practice in recall had a significant impact on later recall. As a result of a single practice-recall session on the same day as the workshops, children at both ages recalled more about the event 4 weeks later in both the repeated and episodic conditions. Children in the repeated condition also reported proportionally fewer generalizations in delayed recall of the first workshop if they had participated in practice recall sessions. It is interesting that practice did not affect the number of intrusions children produced in the repeated condition. Thus, recall practice increased amount but not accuracy of recall.

There are at least three ways in which the practice recall session could have improved later recall. First, in the process of immediately recalling the event, children may have processed some activities more deeply, and those activities may therefore have been more accessible in delayed recall. A second possibility is that the activities cued by the experimenter in the practice session helped to remind children of specific activities they would otherwise have forgotten. However, the finding that practice did not affect the accuracy of children's recall argues against interpreting that memory for specific activities was strengthened. A third possibility is that the questions asked by the experimenter ("Did you sing any songs?" "Did you do any dances?") in the practice session may have provided additional cues to the overall structure of the event. In this case, the practice interview provided children with more structural information about the event that was useful in organizing delayed recall, but did not necessarily minimize intrusions in delayed recall. Thus, practice in remembering events may do more than just review specific episodic details, but may assist the schematization process and thereby improve the exhaustiveness and organization of event memories.

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Publication Practices and Scientific Conduct

The recent disclosures of fraud in the conduct of research, reporting of research, or both in a number of scientific disciplines have prompted a widespread program of self-examination of publication practices and ethics.

The editor joins with APA in reminding authors of the principles of good publication practices and scientific conduct. Prospective authors are directed to the *Publication Manual of the American Psychological Association* (3rd ed.) and to the "Instructions to Authors" printed in this issue. The requirements of data availability, replicability, authorship credit, ethical treatment of subjects, and primary publication of data are important—they are meant to ensure responsible science and appropriate use of scarce and valuable resources.